	GLOBAL ECONOMICS **GRADE: 11TH** **TERM 3 — SOLVED MIDTERM** **HYPOTHESIS TEST DESIGN AND DECISION (C1–C7)** **TEACHER'S NAME: Nicolás López Cuéllar**	**THIRD TERM 2025–2026**

Learning objective: Design and apply hypothesis tests from real contexts by identifying the parameter, writing hypotheses, selecting the test direction, applying a proportion test and a mean test, interpreting statistical errors, and determining the minimum sample size needed to reject the null hypothesis.	**Evaluated criteria:** C1: Describes the concept of a statistical hypothesis. C2: Identifies the null and alternative hypotheses in a statistical test. C3: Determines whether a test should be one-tailed or two-tailed. C4: Applies a hypothesis test in a population proportion setting. C5: Applies a hypothesis test in a population mean counting setting. C6: Identifies Type I or Type II error and its consequences. C7: Finds the minimum sample size needed to reject H_0.

Critical values used

$$z_{0.95} = 1.645 \qquad z_{0.975} = 1.960 \qquad z_{0.99} = 2.326$$

For right-tailed tests, reject when $z > z_{\text{critical}}$. For left-tailed tests, reject when $z < -z_{\text{critical}}$. For two-tailed tests, reject when $|z| > z_{\text{critical}}$.

Problem 1 — Grade 11 punctuality audit

The school compares punctuality between Prom 2025 and Prom 2026. The variable is the number of late arrivals to first class per morning.

$$1 = \text{more than 9 late arrivals} \qquad 0 = \text{not more than 9 late arrivals}$$

#	Prom 2025	2025-bin	Prom 2026	2026-bin
1	5	0	5	0
2	6	0	7	0
3	7	0	8	0
4	8	0	9	0
5	9	0	10	1
6	4	0	5	0
7	6	0	8	0
8	7	0	9	0
9	10	1	11	1
10	5	0	6	0
11	8	0	10	1
12	7	0	8	0
13	6	0	7	0
14	9	0	10	1
15	11	1	12	1

#	Prom 2025	2025-bin	Prom 2026	2026-bin
16	5	0	6	0
17	7	0	8	0
18	8	0	10	1
19	10	1	11	1
20	6	0	7	0
21	10	1	10	1
22	7	0	8	0
23	6	0	7	0
24	8	0	10	1
25	11	1	12	1
26	5	0	6	0
27	7	0	8	0
28	9	0	10	1
29	12	1	11	1
30	10	1	9	0

Summary data:

$$n_{25} = 30 \qquad \bar{x}_{25} = 7.63 \qquad \sigma_{25} = 2.09 \qquad x_{25} = 7 \qquad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$n_{26} = 30 \qquad \bar{x}_{26} = 8.60 \qquad \sigma_{26} = 3.24 \qquad x_{26} = 12 \qquad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Use $\alpha = 0.05$

C1

Mean setting

μ_{25} = true mean late arrivals for Prom 2025 $\quad$ μ_{26} = true mean late arrivals for Prom 2026

$$n_{25} = 30 \qquad \bar{x}_{25} = 7.63 \qquad \sigma_{25} = 3.40$$
$$n_{26} = 30 \qquad \bar{x}_{26} = 8.60 \qquad \sigma_{26} = 3.24$$

Proportion setting

p_{25} = true high-lateness proportion for Prom 2025 $\quad$ p_{26} = true high-lateness proportion for Prom 2026

$$x_{25} = 7 \qquad n_{25} = 30 \qquad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$
$$x_{26} = 12 \qquad n_{26} = 30 \qquad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Using Prom 2025 as benchmark

C2

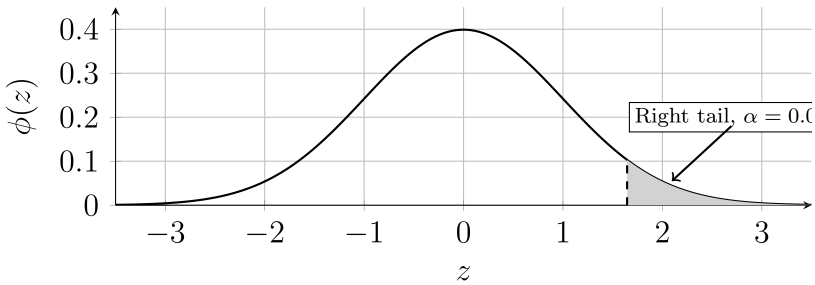
Benchmark direction: Prom 2026 vs Prom 2025

$$H_0\colon p_{26} = 0.2333 \qquad H_a\colon p_{26} > 0.2333 \qquad p_0 = 0.2333$$
$$H_0\colon \mu_{26} = 7.63 \qquad H_a\colon \mu_{26} > 7.63 \qquad \mu_0 = 7.63$$

C3

Right-tailed

$$H_a\colon p_{26} > 0.2333 \qquad H_a\colon \mu_{26} > 7.63$$



C4

$$\hat{p}_{25} = 0.2333 \qquad \hat{p}_{26} = 0.40 \qquad n_{26} = 30 \qquad k = 2.20$$
$$z = \frac{\hat{p}_{26} - \hat{p}_{25}}{\sqrt{k\,\frac{\hat{p}_{25}(1-\hat{p}_{25})}{n_{26}}}} = \frac{0.40 - 0.2333}{\sqrt{2.20\,\frac{0.2333(0.7667)}{30}}} \approx 1.46$$
$$z_{\text{critical}} = 1.645 \qquad 1.46 < 1.645$$

Fail to reject H_0. $p_{26} > 0.2333$ is not supported.

C5

$$\bar{x}_{25} = 7.63 \qquad \bar{x}_{26} = 8.60 \qquad \sigma_{26} = 3.24 \qquad n_{26} = 30$$
$$z = \frac{\bar{x}_{26} - \bar{x}_{25}}{\frac{\sigma_{26}}{\sqrt{n_{26}}}} = \frac{8.60 - 7.63}{\frac{3.24}{\sqrt{30}}} \approx 1.64$$
$$z_{\text{critical}} = 1.645 \qquad 1.64 < 1.645$$

Fail to reject H_0. $\mu_{26} > 7.63$ is not supported.

C7

Proportion test

$$\frac{0.1667}{\sqrt{2.20\,\frac{0.2333(0.7667)}{n_{26}}}} > 1.645$$

$$n_{26} > \left(\frac{1.645\sqrt{2.20 \cdot 0.2333(0.7667)}}{0.1667}\right)^2 \approx 38.34$$

$$n_{26,\text{min}} = 39$$

Mean test

$$\frac{0.97}{\frac{3.24}{\sqrt{n_{26}}}} > 1.645 \Rightarrow n_{26} > \left(\frac{1.645(3.24)}{0.97}\right)^2 \approx 30.18$$

$$n_{26,\text{min}} = 31$$

C6

Initial sample size $n_{26} = 30$: **Fail to reject H_0.** both tests do not support increases.

$$z_p = \frac{0.1667}{\sqrt{2.20\,\frac{0.2333(0.7667)}{39}}} \approx 1.66 > 1.645$$

$$z_\mu = \frac{0.97}{\frac{3.24}{\sqrt{31}}} \approx 1.67 > 1.645$$

Minimum sample sizes: **Reject H_0.** both tests support increases. Type II error possible when failing to reject; Type I error possible when rejecting.

Using Prom 2026 as benchmark

C2

Benchmark direction: Prom 2025 vs Prom 2026

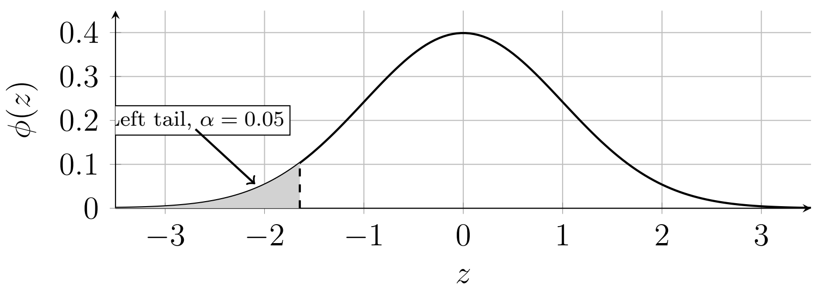
$$H_0\colon p_{25} = 0.40 \qquad H_a\colon p_{25} < 0.40 \qquad p_0 = 0.40$$

$$H_0\colon \mu_{25} = 8.60 \qquad H_a\colon \mu_{25} < 8.60 \qquad \mu_0 = 8.60$$

C3

Left-tailed

$$H_a\colon p_{25} < 0.40 \qquad H_a\colon \mu_{25} < 8.60$$



C4

$$\hat{p}_{26} = 0.40 \qquad \hat{p}_{25} = 0.2333 \qquad n_{25} = 30 \qquad k = 1.50$$

$$z = \frac{\hat{p}_{25} - \hat{p}_{26}}{\sqrt{1.50\,\frac{\hat{p}_{26}(1-\hat{p}_{26})}{n_{25}}}} = \frac{0.2333 - 0.40}{\sqrt{1.50\,\frac{0.40(0.60)}{30}}} \approx -1.52$$

$$z_{\text{critical}} = -1.645 \qquad -1.52 > -1.645$$

Fail to reject H_0. $p_{25} < 0.40$ is not supported.

C5

$$\bar{x}_{26} = 8.60 \qquad \bar{x}_{25} = 7.63 \qquad \sigma_{25} = 3.40 \qquad n_{25} = 30$$

$$z = \frac{\bar{x}_{25} - \bar{x}_{26}}{\frac{\sigma_{25}}{\sqrt{n_{25}}}} = \frac{7.63 - 8.60}{\frac{3.40}{\sqrt{30}}} \approx -1.56$$

$$z_{\text{critical}} = -1.645 \qquad -1.56 > -1.645$$

Fail to reject H_0. $\mu_{25} < 8.60$ is not supported.

C7

Proportion test

$$\frac{0.1667}{\sqrt{1.50\,\frac{0.40(0.60)}{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645\sqrt{1.50 \cdot 0.40(0.60)}}{0.1667}\right)^2 \approx 44.54$$

$$n_{25,\text{min}} = 45$$

Mean test

$$\frac{0.97}{\frac{3.40}{\sqrt{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645(3.40)}{0.97}\right)^2 \approx 33.23$$

$$n_{25,\text{min}} = 34$$

C6

Initial sample size $n_{25} = 30$: **Fail to reject H_0.** both tests do not support decreases.

$$z_p = \frac{-0.1667}{\sqrt{1.50\,\frac{0.40(0.60)}{45}}} \approx -1.66 < -1.645$$

$$z_\mu = \frac{-0.97}{\frac{3.40}{\sqrt{34}}} \approx -1.66 < -1.645$$

Minimum sample sizes: **Reject H_0.** both tests support decreases. Type II error possible when failing to reject; Type I error possible when rejecting.